UNIVERSITY OF CAMBRIDGE

DARTFormer: Finding The Best Type Of Attention

Jason Ross Brown[1], Yiren Zhao[2], Ilia Shumailov[3], Robert D. Mullins[4]

[1]jrb239@cam.ac.uk [2]a.zhao@imperial.ac.uk [3]ilia.shumailov@chch.ox.ac.uk [4]robert.mullins@cl.cam.ac.uk

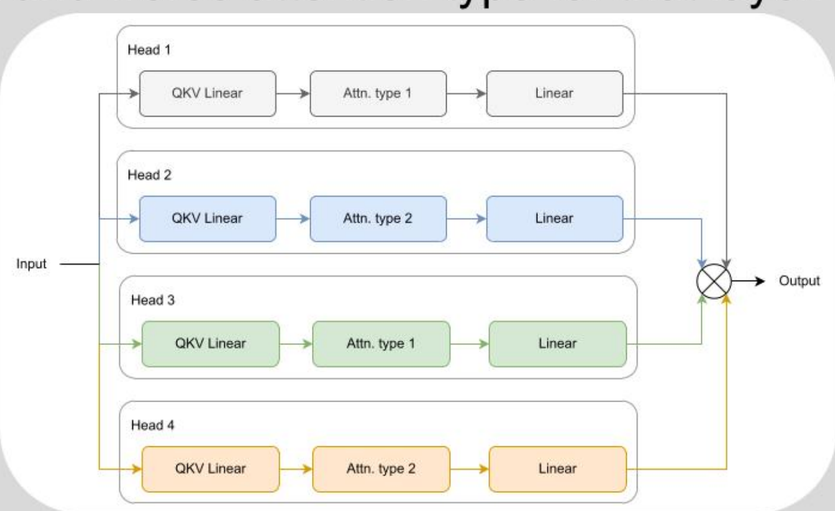
Motivation

- There is a wide range of efficient Transformer attention mechanisms whose relative performance often depends on the task
- Different attention heads are supposed to pick up different patterns
- DARTS-like neural architecture search is highly effective at finding good network topologies

DARTFormer

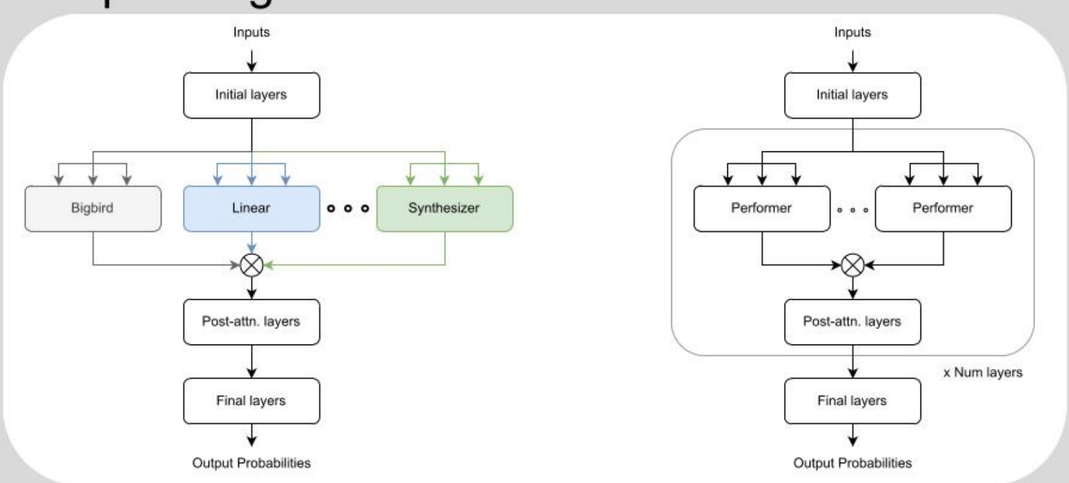
Attention Search

- *Idea*: Can we apply a DARTS-like method to identify good and bad attention mechanisms?
- By training a supernetwork with heads that have different attention types in each layer, we can then use masked validation to identify the current best and worse attention type for that layer.



Optimal Homogeneous Attention

Applying the attention search idea to a single layer Transformer and picking the best performing attention, we can hope to identify the best attention for that task when operating in a full Transformer model.



Optimal Heterogenous Attention

- *Method 1:* Attention search on a single layer pruning the worst heads until we have the right size for a single layer. Stack this layer and retrain from scratch.
- *Method 2:* Attention search on a single layer and then take the top 4 scoring mechanisms. Stack a uniform mix and retrain from scratch.
- *Method 3:* Attention search across many layers, picking the best attention in each one. Retrain from scratch with possible different attention in each layer

Experimental Results

- Homogeneous search works on 2/3 tasks

Attention type	Text classification		Listops		Document Matching	
	Acc	Score	Acc	Score	Acc	Score
Bigbird	*62.7*	*0.30*	36.7	0.25	63.9	*0.011*
Linear Transformer	64.4	0.17	37.1	-0.40	64.2	-0.006
Linformer	58.6	-0.28	29.9	0.00	*64.5*	0.006
Local	56.3	0.14	36.8	0.30	58.0	0.000
Longformer	61.8	-0.20	36.8	*0.40*	61.2	**0.039**
Performer	**64.5**	**0.51**	36.2	0.30	65.0	0.000
Reformer	55.7	0.44	**37.8**	**11.85**	58.3	0.028
Sparse Transformer	61.3	0.02	*37.0*	0.50	63.2	0.000
Synthesizer	61.4	-0.03	36.5	0.30	**71.1**	-0.006

- Heterogeneous attention fails to outperform the best homogeneous models

Task	Best Homogeneous	Method 1	Method 2	Method 3
Token Level Text Classification	**65.9**	-	-	64.6
Byte Level Text Classification	**64.5**	63.9	64.4	-
Listops	**37.8**	36.3	37.1	-
Document Matching	**71.1**	67.0	64.7	-

Conclusion

- A DARTS-like method can often be used to find the best attention for a task by training only a single layer
- Searching for and using heterogeneous 'mixed attention' Transformers does not give good results.
- Perhaps different mechanisms do not provide as large functional biases, or the biases are less useful for learning good representations in the attention layer than was previously thought